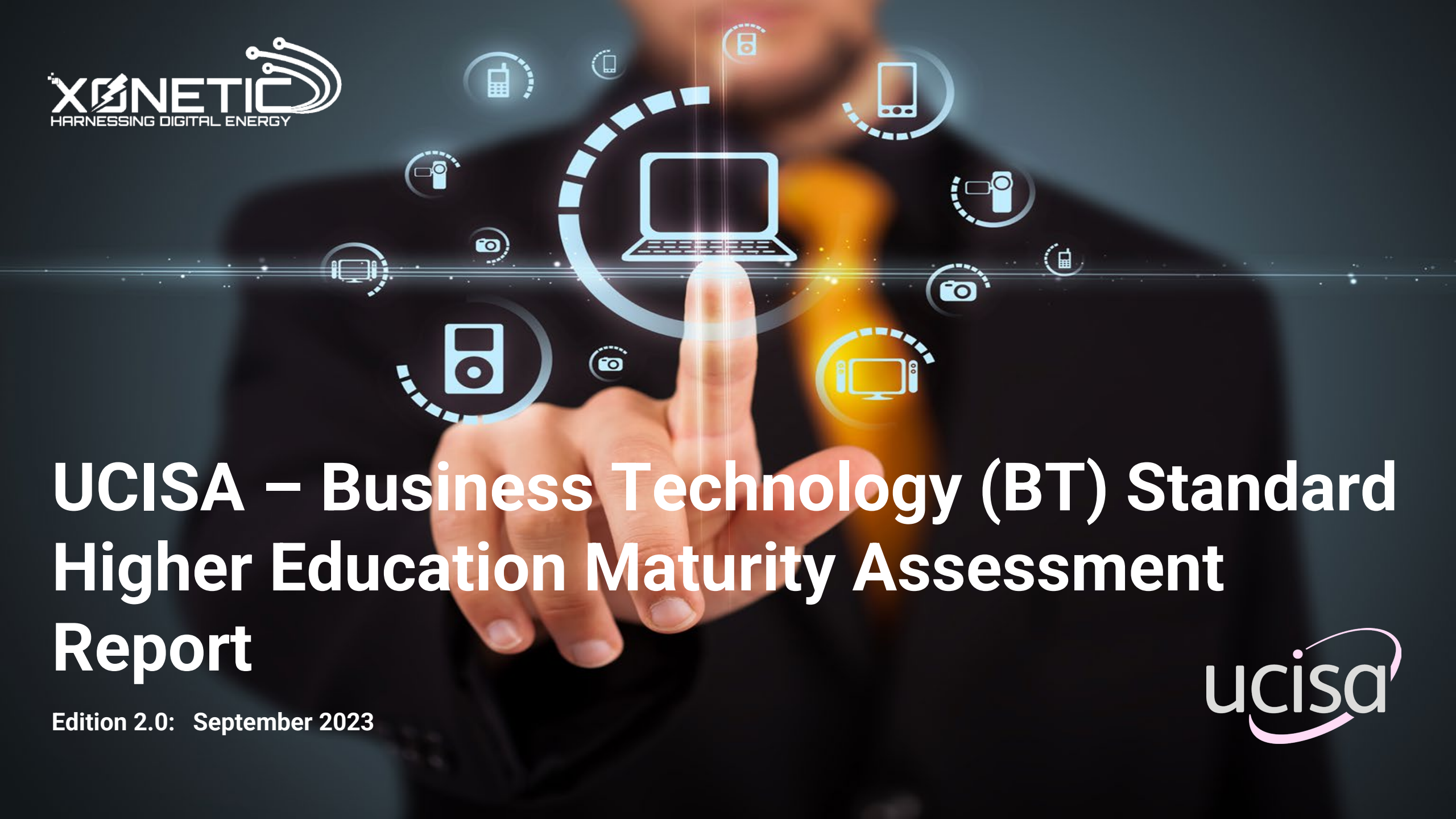

XONETIC
HARNESSING DIGITAL ENERGY

UCISA – Business Technology (BT) Standard Higher Education Maturity Assessment Report

Edition 2.0: September 2023

ucisa

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one FORWARD

Welcome to the second edition technology maturity assessment report for the Higher Education (HE) sector which has been sponsored by UCISA in partnership with Xonetic who are the sole **UKI partner** for the **Business Technology Standard**.

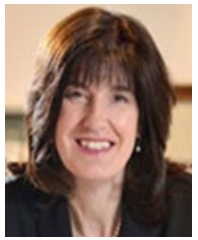
The report represents the findings from fourteen institutions who have taken part over the last 18 months. Over time, we hope to keep the overall dataset updated and refreshed on a periodic basis to drive an enhanced sector view, as both new and existing institutions choose to participate.

We would like to thank the institutions and respondents who have participated in this maturity assessment process. It has been performed against the **Business Technology (BT) Standard Capability Model,** which provides a framework to assess digital readiness.

The report highlights a deepening acceptance and usage of the BT Standard across the sector.

Insights created in the last report have broadly speaking, held, underpinned by the additional universities that chose to take part in this year's assessments.

These insights have been highly valued by participating universities, especially set against the senior technology stakeholder challenges from 2022 and moving into 2023. We hope that you find this report an insightful and interesting aid to promote discussion about your digital transformation. Please reach out to us to learn how your institution can get involved!



Deborah Green
CEO
UCISA

Nick Russell
Managing Director
Xonetic Ltd

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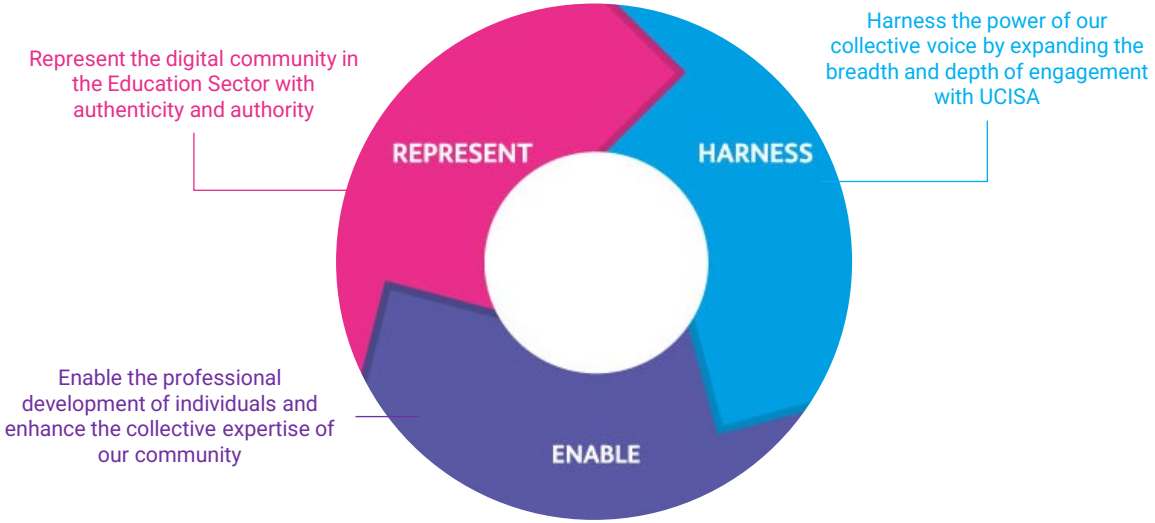
ABOUT UCISA

Founded in 1993, UCISA is the member-led professional body for digital practitioners in education. Open and inclusive, we connect our communities of practice and key stakeholders, working together to share our collective technical knowledge and digital expertise to help transform teaching, learning and research by enabling operational efficiency and an excellent student experience.

Our **mission** is to connect, share and transform. and our **goals** are to :



Almost all UK Higher Education Institutions are UCISA members along with a growing number of Further Education and Sixth Form Colleges. Institutional membership automatically grants the benefits of the UCISA community to all staff working within the institution.

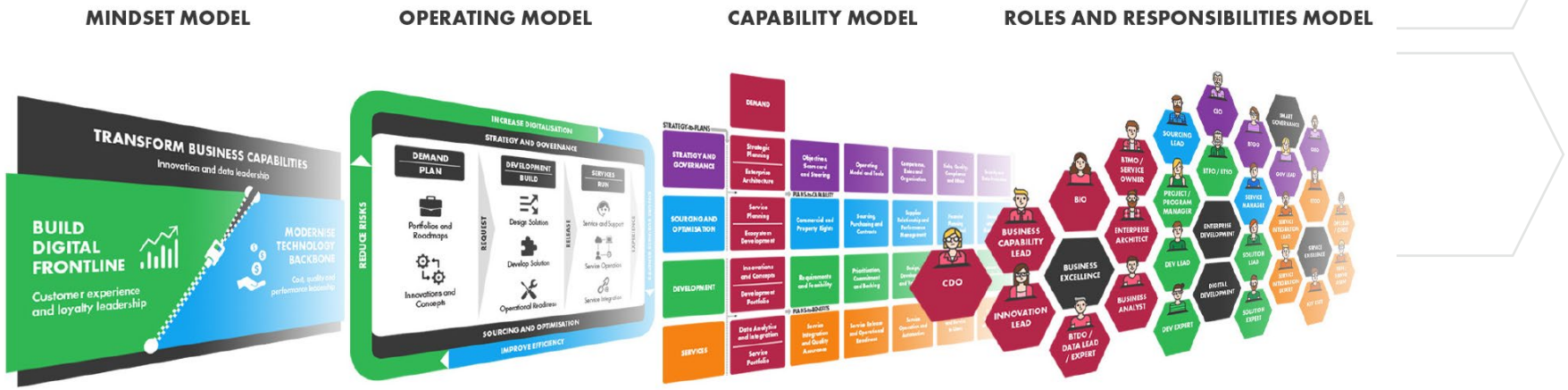
Our community collaborates, networks, and shares their inspirational thinking, practices, and procedures so that we can all learn more, advance faster and be more effective. We also work closely with technology and digital service suppliers to the sector – helping them better understand the sector's requirements so that they can deliver the solutions needed within member institutions.

Our **vision** is to:

We ensure that institutional members' staff at all stages of their career have access to professional development to aid their career progression. We develop training and skills for the future – horizon scanning to explore what our community needs. We draw on and harness the expertise of our corporate members, collaborating and encouraging co-creation and innovation to develop products, services and support that meet the current and future needs of an ambitious education sector. We empower our membership and lead by example.

three

WHAT IS THE BUSINESS TECHNOLOGY STANDARD



The **Business Technology (BT) Standard** is an industry best practice and open-source management framework to plan, build and run information technology in today's technology-driven business world.

The BT Standard creates value for business by:

- Enabling a unified understanding and common language between business and IT departments
- Instilling best practices, models and tools
- Improving opportunities to make the most out of digital innovation
- Being relevant for small, medium and large companies

The BT Standard provides a unified management model overlaying different technology management areas. It binds together specific management models such as Scaled Agile Framework (SAFe) and DevOps for digital development and ITIL and IT4IT for service management in new and innovative ways.

The BT Standard is ideal for any organisation starting or going through a digital transformation and is used in hundreds of companies across Europe.

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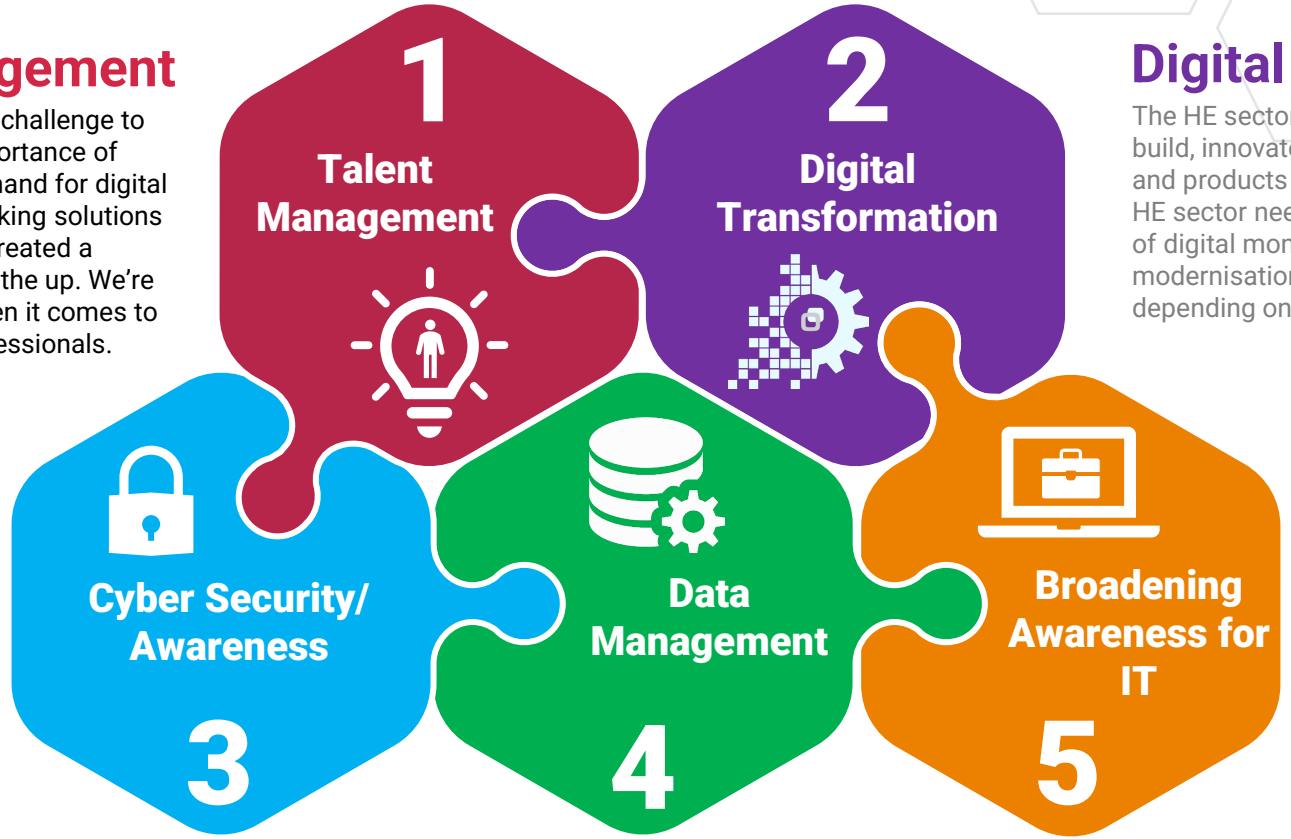
DIGITAL LEADERSHIP CHALLENGES

The top 5 Challenges for Leaders in the Higher Education sector for 2023, continued from 2022, drawn from feedback from UCISA members

Talent Management

Talent management is a continued challenge to institutions with the increasing importance of technology, combined with the demand for digital solutions. The need for remote working solutions over the past couple of years has created a domino effect – skill demand is on the up. We're seeing a perfect storm of sorts when it comes to finding and retaining skilled IT professionals.



Digital Transformation

The HE sector is establishing new/enhanced ways to build, innovate and deliver their institutions services and products digitally to drive service excellence. The HE sector needs to build a foundation for the creation of digital momentum or energy, be that of modernisation or overall strategic transformation depending on what is right for each institution.

Cyber Security/ Awareness

The continued risk of Cyber attacks across the sector remains a highly visible priority. Implementation of the right capability (people/technology) and implementation of awareness across institiutions is vital. It's important to educate everyone, academics, professional support staff and students regularly on cybersecurity risks and how to avoid them.

Broadening awareness for IT

Understanding of IT themes is essential for digital leaders across all areas of a business. It is now no more acceptable for business leaders to say they do not understand IT (key cyber risks for example) than it would be to say they cannot understand finance (read a balance sheet for example). Business Technology awareness across the institutional top table is increasingly important.

Data Management

Data is a strategic business asset and one of the most valuable resources an institution has. But its value is dependent on quality, relevance and scope. Turning data into a high value commodity requires a journey of transformation (people/culture, process/governance and technology). As an institutions' level of data management matures so the benefits of data grows too.

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INTRODUCTION

How the assessment process works

UCISA and Xonetic have worked together to support the continued adoption of the rapid technology maturity assessment for Higher Education institutions based on the **Business Technology (BT) Standard.**

The assessment is performed across three stages, the first focusing on a non-qualitative survey approach across technology within the institution (business and IT respondents) and secondly making an assessment across all 28 areas within the **Business Technology Capability Model**. The third focuses on three identified capability areas per institution that warrant further investigation. The assessment process is performed over a 3-week period using a virtual tool. Just **15 minutes** are needed per day across respondents. Formal individual participant interviews are not required, although opportunity for individuals to provide feedback within the tool is available. This can provide additional context to the assessment insights.

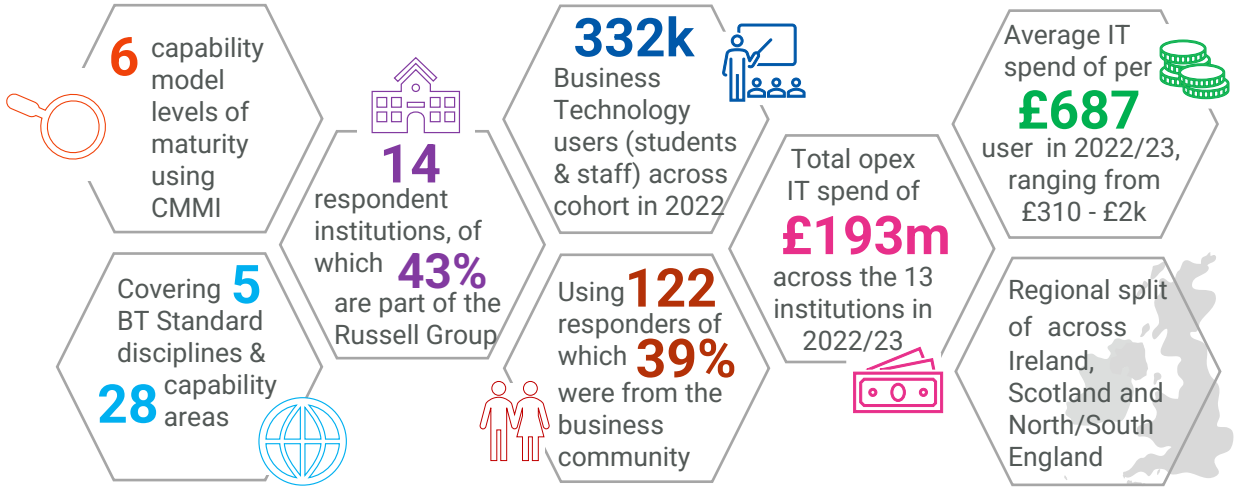
For this second iteration, the report is now based on fourteen institutions ranging from ancient (pre 1900) to post 1992 (previously called Polytechnics) and **over one hundred & twenty** responders from both IT and the wider organisation, with knowledge of the way technology is used within their institution. This provides a c'10% sample size for related oberservations and opportunities representing the HE sector. We ask for your continued support in the maturity assessment process[1] during 2023, further participants will allow the data to become more robust, and provide even more insight.

[1]Please refer to Appendix 2 for further details on the assessment process and commercials.

This report does provide insights and observations from the data analysis across the assessment institutions, aligned to the BT Standard disciplines. In addition, the analysis identified opportunities for shared improvements in the business value of technology. These opportunities will continue to be discussed within the UCISA Business Technology Community of Practice, which is the hub of the community within Higher Education (HE).

Conclusions can be taken from the data, which have been presented in this report, in order to promote discussion across the sector. External influences also play a factor in our observations, in particular the need for greater digital transformation in the economy as a whole. The challenges of remote working and touch economic conditions continue to bring strong headwinds which are common to all institutions.

Assessment demographics



> The BT Standard supports institutional digital transformation within and beyond IT. It aligns and integrates multiple tried and tested good practice frameworks, so we don't have to align and integrate them ourselves.
>
> **Nick Leake** (CIO)
> King's College London

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INTRODUCTION

Higher Education Reference Model (HERM) and the Business Technology Standard

The Higher Education Reference Model (HERM) is curated and extended by warm collaboration between CAUDIT, UCISA, EDUCAUSE and EUNIS. HERM is now used by more than four-hundred universities throughout the world and provides the standardised business and data architectures that communicate a generalised view of how higher education institutions are organised and the information they use.

The individual business enabling capabilities can be thought of as the 'building blocks' required to run a UK HE organisation. It supports technology alignment to the business side of the institution based on the value chains/streams that keep the institution running e.g.: Strategy Management, Business Capability Management, Governance, Risk & Compliance, Information Management, Financial Management, etc…

As part of the assessment process and this report we did not actively consider the HERM due to the compact timeframes, but we can see potential synergies against the **BT Standard** disciplines and the associated 28 standardised capability areas that can be used to guide an organisations business technology in a holistic way.

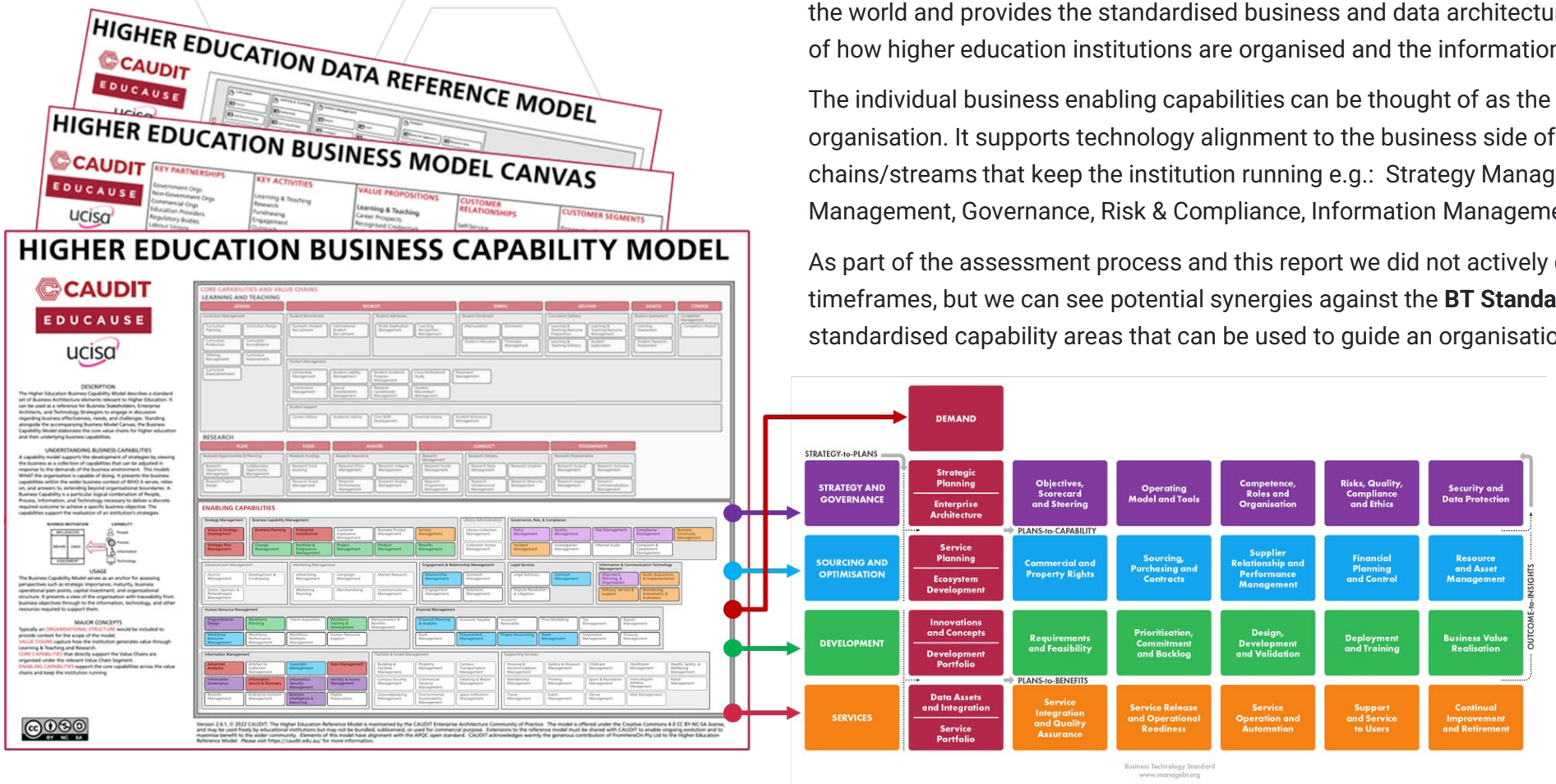


Figure 1 – HE Reference Model (v2.61) Alignment to the BT Standard

There is further opportunity across the UCISA community to assess how the HERM and the BT Standard capability frameworks can operate together to drive a continued qualitative assessment approach against each building blocks using a standard capability maturity (CMMI) scale (see Appendix 1).

six EXECUTIVE SUMMARY

The assessment indicated that each discipline averaged at **level 2 (Managed)**, with an average cohort maturity of **level 2.27 (Managed)** across all disciplines. The assessment data provides a range between the minimum and maximum maturity and standard deviation (variability of the data values from the mean (average)) for each discipline.

Xonetic has suggested a stretch target for the sector at this point of level **3.0 (Defined).** There is no suggestion of institutions performing poorly, this target has been set purely to promote discussion. It is an expected and correct finding that maturity varies as institutions vary in type, size and complexity, necessarily driving a spread of IT capabilities dependant on need.

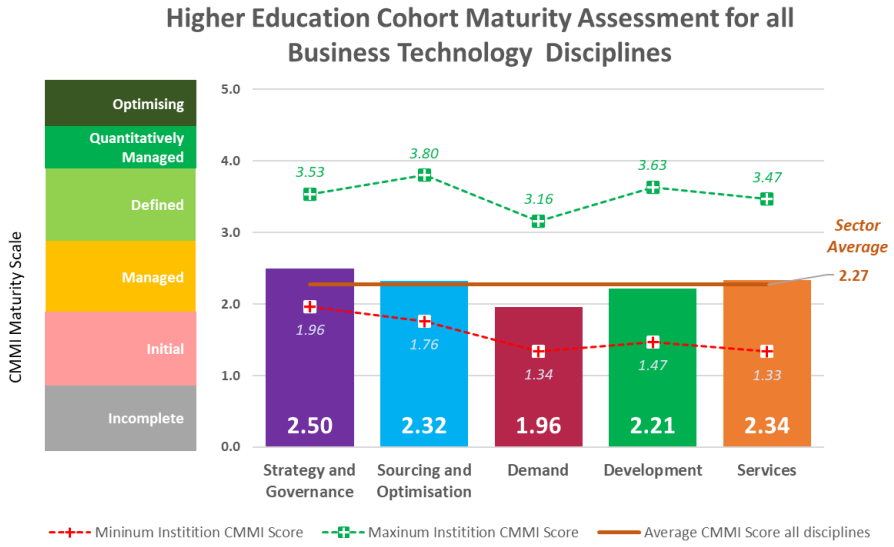


Figure 2 – HE Sector BT Standard Discipline Maturity Assessment[3]

Indeed, as the number of respondents increases, we see a larger spread of maturity levels across the different institutional types with post 1992 institutions having the highest level of maturity.

The average value here is therefore a rough approximation of overall sector maturity.

As more institutions take part the enhanced dataset is useful to drive more discussion across the UCISA community.

Initial assessment findings

HE average CMMI Level is **2 MANAGED**

Lowest average maturity of **1.96** **INITIAL** for the Demand discipline

Post 1992 institutions have the highest maturity

Highest average maturity of **2.5** for the Strategy & Governance discipline

Institutions stretch target of Level **3** **DEFINED** or higher[2]

Only **14%** of institutions have a maturity above level **3** **DEFINED**

> The Assessment report provides a clear tool to articulate the capability areas for improvement as part of our IT strategic planning
>
> **Mark Johnston** (Director of IT)
> University of Glasgow

[2] This needs to be assessed on a per capability basis, as appropriate for that institution

[3] Participants assessed the Business Technology Standard Capability Model using the CMMI levels of Incomplete, Initial, Managed, Quantitatively Managed and Optimising

six EXECUTIVE SUMMARY

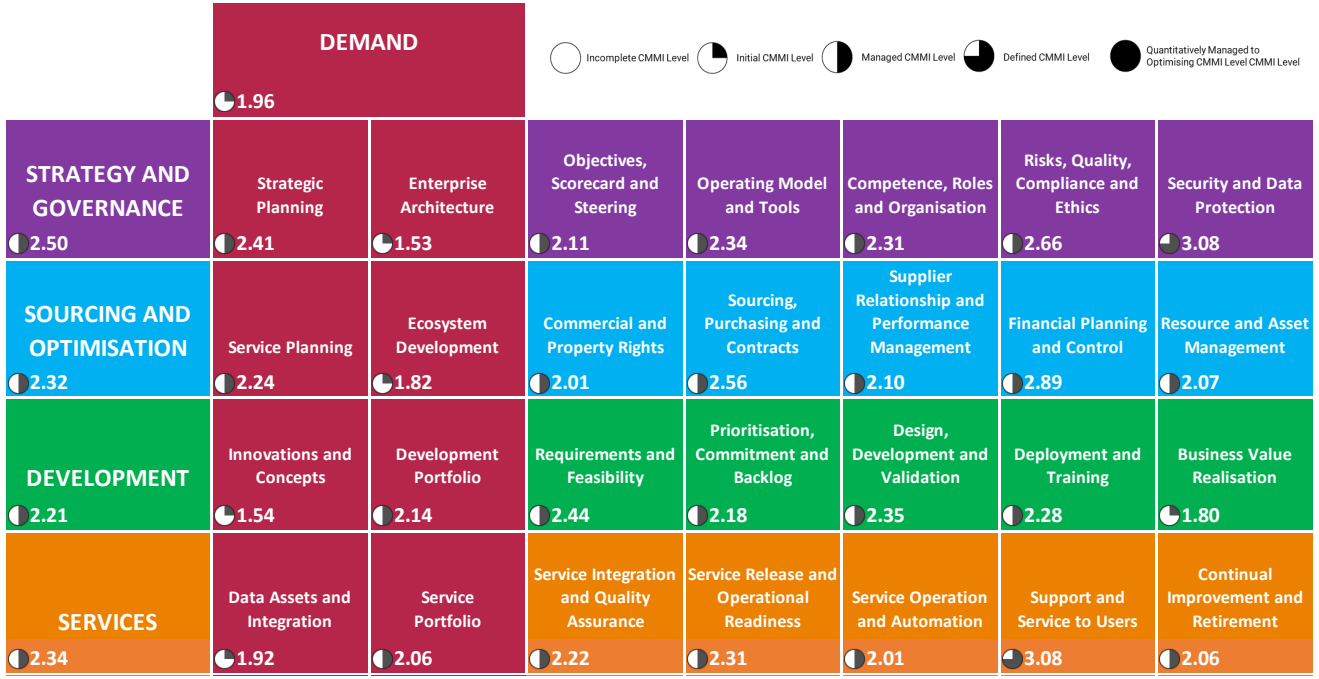


Figure 3 – HE Sector BT Standard Capability Maturity Assessment Heatmap

> The BT Standard professionalises and standardises the way IT, Professional Service functions, faculties and suppliers are organised and run. This delivers more value, more quickly to students and staff and provides a better working environment for everybody
>
> **Nick Leake** (CIO)
> King's College London

What have we observed

By using a maturity assessment heatmap set against the Business Technology Capability Model we are able to highlight potential focal areas.

- The **Strategy & Governance**, **Sourcing & Optimising**, **Development** and **Services** disciplines all show **level 2** (**Managed**);
- The **Demand** discipline has the lowest overall maturity at **level 1 (Initial)** with 64% of institutions operating at this level;
- There was a higher level of maturity around "Support and Service to Users" with 71% of the institutions measuring **level 3 (Defined)** or above, with the highest institution at **level 4.33 (Quantitatively Managed)**; and
- this was followed by 64% of institutions "Security & Data Protection" measuring at **level 3 (Defined)** or above.

Across the capability disciplines the cohort participants indicated:

- 2 capabilities at level Defined CMMI level
- 21 capabilities at level Managed CMMI level
- 5 Capabilities at level Initial CMMI level

Five Current State Observations

There are a number of key observations across the BT Standard capability disciplines. These themes of improvement across the institutions can be taken forward, both at an individual institution level or as a collective under the UCISA community of practice. The following provides the top five (5) observations :

#01 – Demand discipline improvement

The **Demand discipline** has the lowest maturity across the institutional assessment cohort. This is likely to **impede the institution's ability to drive their digital transformation**. Institutions should consider reviewing their governance structures and related roles, responsibilities and competencies. Example improvements would be digital strategy/planning (**digital runway**) and integration across Demand, Development and Services.

Many organisations are now considering how to rebalance this picture, as the challenge shifts from providing the best IT support service to a team that can innovate and drive business value. Improvements in tooling and automation/AI are essential to prevent any loss in service provision if organisation attention is turned to strengthen Demand & Development.

See Figure 4 which introduces the question of organisation balance?

Note: The numbers shown here are not a suggested benchmark, they are thematic to promote further discussion.

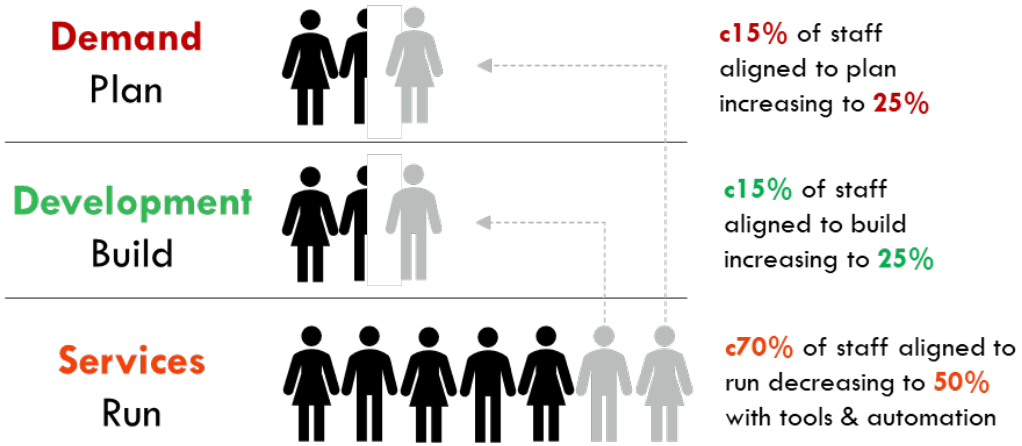


Figure 4 – What is the right balance of resource?

EXECUTIVE SUMMARY

Define institution business domains & Value Streams

Institutional adoption of a BT Capability model

BT Standard Capability model potential alignment to HE Reference Model

Institutional build of defined BT roles & responsibilities

#02 – BT Standard Capability Model Adoption

The **BT Standard Maturity Assessment** has now been taken by fourteen institutions which highlights an improving picture for a common approach to IT Management across the sector.

All institutions who have taken part have indicated the high value and usefulness that the assessment has provided to both IT and Business teams alike looking to understand the complex nature of IT provision. The assessment has provided useful focus for these organisations and helped to shape their IT Management direction.

However further adoption of the standard remains limited in breadth and depth, meaning there remains a large opportunity ahead. UCISA continue to play an important role, along with Xonetic, in continuing to look for common and cost-conscious opportunities to further embed the BT across the sector.

The use of the **BT Standard Capability Model**, and the defined roles and responsibilities across capabilities for all institutions as part of the wider **BT Standard** adoption, is a starting point to drive consistency of how business technology is delivered and measured across the sector. Coupled with the potential synergy we have seen (please refer to Introduction) with the BT Standard Capability and HE Reference models – this creates a foundation by which the sector can operate for digital transformation.

#03 – Data Management

The maturity of **data management** remains low with 43% of institutions in the assessment scoring at level **1-Initial** maturity, and the remaining only reaching low level **2-Managed** maturity, which suggests processes are poorly managed or controlled with unpredictable outcomes and unstructured approaches in use - Low quality and high risk.

A lack of overarching Institutional Data Strategy, and related domain data strategies prevents turning data into a high value commodity.

Data as an asset – can drive business value, help make informed decisions, improve operational excellence, support monetisation of data and enhance user experiences (academics, staff and students).

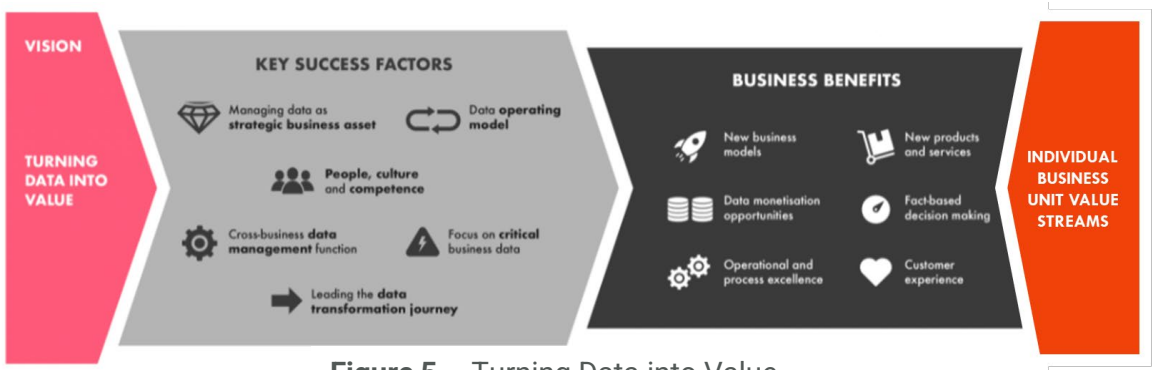
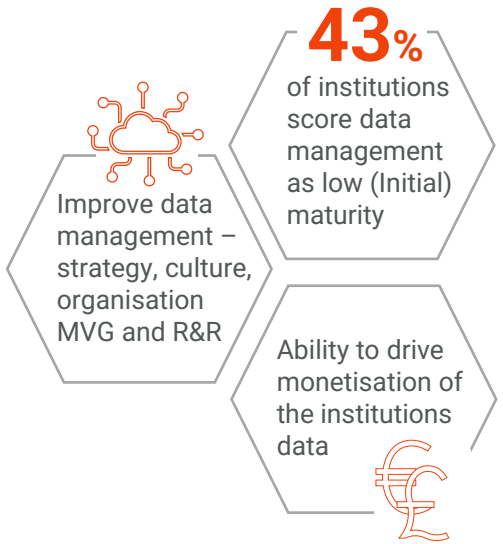


Figure 5 – Turning Data into Value

Although the journey will start with a data strategy, institutions will need to develop the following:

- clear understanding of institutional **data/data** flows and **data integration** capabilities against these flows;
- a data capability model for planning, building and running data products and data services that helps define the roles needed across the institution aligned to **value streams**;
- **roles, responsibilities** and **governance** across business & technical domains. BT standard roles and structures can be used as a guide;
- well managed and maintained **data quality**; and
- An **organisation** and **culture** which understands the importance of data.



EXECUTIVE SUMMARY

#04 – Enterprise Architecture Capability Planning

Enterprise Architecture (EA) capability was also seen as an area of concern across the institutional cohort with 10 (71%) of institutions in the assessment scoring at level **1-Initial** maturity.

We are aware of the work of the UCISA Enterprise Architecture Group, which launched in global Higher Education Reference Model (HERM) earlier this year and which provides collaboration and support across the UK HE Enterprise Architecture community and would encourage members to size the opportunity to further develop/enhance their institutions Enterprise Architecture capability.

The EA function must be the facilitator of capability planning and design within an organisation, and to this end must be involved (with varying levels of intensity) in the entire change journey. We see in the broader marketplace that EA teams have become disconnected from business and IT teams. There is no suggestion that this has occurred within the institutions that have been assessed however the creation of value streams and increased digital transformation forces evolution of practices and governance models.

In the Business Technology Standard, we propose a capability based planning approach, placing ecosystem planning in a central role. The modularity of this approach allows for a more agile development of architecture for different business areas.

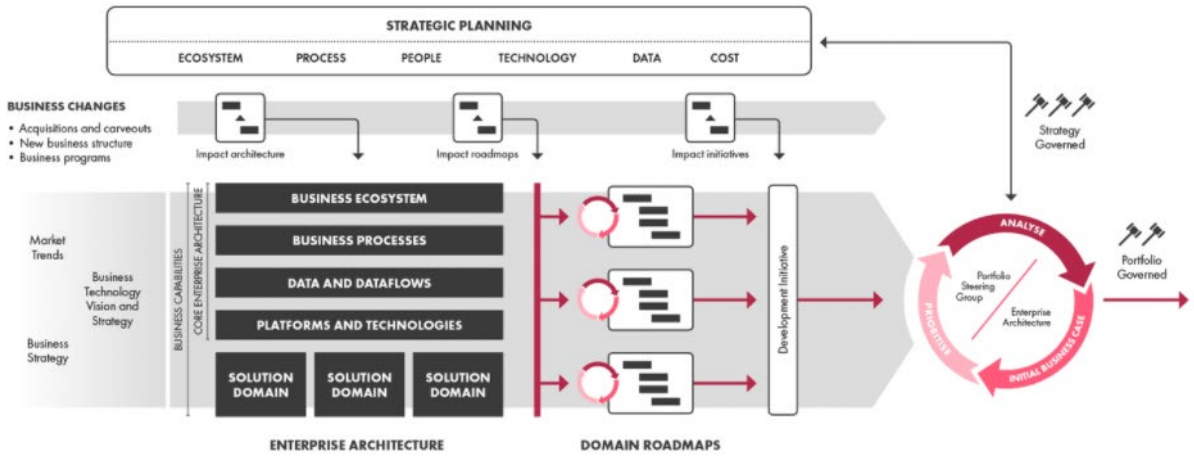
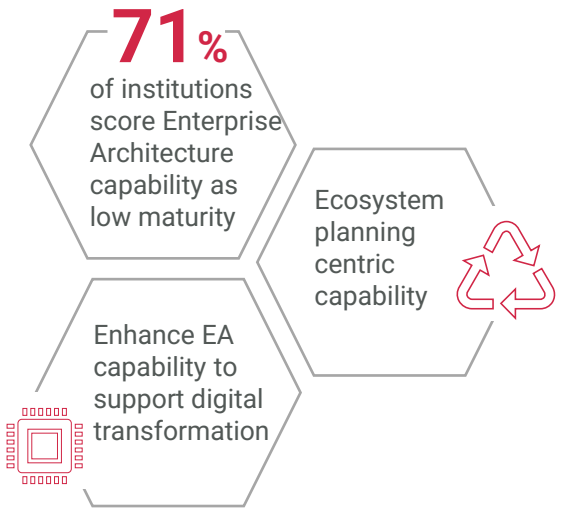


Figure 6 – The enterprise architecture centric capability planning

The model is compliant with traditional enterprise architecture by The Open Group Architecture Framework (TOGAF) encompassing four aspects of architecture: **business**, **information**, **application** and **technology** architectures



#05 – end-2-end delivery of business value

We have seen a combination of BT Standard capabilities with low maturity which underpin the delivery of business value – **Innovation & Concepts, Business Value Realisation** and **Ecosystem Development.** These when combined provide a maturity level of **1-Initial** across all institutions.

The starting challenge is the overall adoption of the **value stream** concept as part of the operating model. The goal is to provide end-to-end objectives that create business value across the institution business domains. Value streams should cross business, technology or other domain focus for value creation.

Once the foundation of value streams are place there will be a need to review and enhance the benefits realisation process and implementation of minimal viable governance (which increases the speed of change).

The fundamental objective is to ensure that all activity contributes measurable value, and that a customer centric culture is fostered within the organisation. This helps to ensure the success of digital transformation initiatives, which are often closely intertwined with broader organisation success criteria/ambition.

The use of supplier partnerships, especially when driving an institution digital transformation, will be critical and should start with a clear sourcing strategy/vision – **RIGHT** Sourcing.

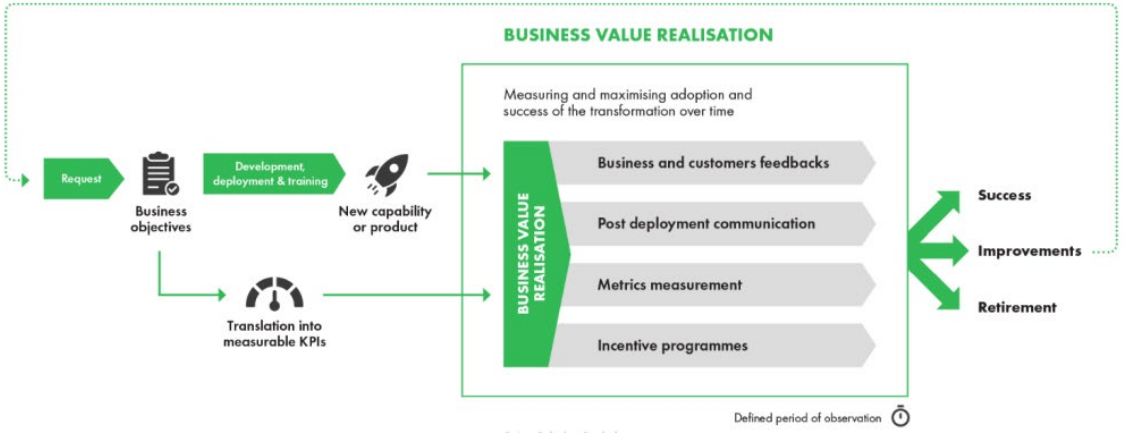


Figure 7 – Business Value Realisation

Development of both sector and technology-based ecosystems will support value creation at a sector and institution business domain level.

71% of institutions score Ecosystem Development at level **1-Initial** maturity

86% of institutions score Innovation & Concepts at level **1-Initial** maturity or lower

64% of institutions score Business Value Realisation at level **1-Initial** maturity

Enhanced change delivery framework to improve stage gates

Sector and technology Ecosystem development to support digital transformation

seven

CONCLUSION

The Business Technology maturity assessment for Higher Education institutions has identified a number of key observations against the BT Standard disciplines and capability areas that we believe should be focus of analysis and priority for those institutions who have participated:

#01 Focus on the development of the Demand discipline related capabilities to create appropriate strategy and planning to support digital transformation.

#02 Institutional wide adoption of the Business Technology Standard and related R&R, and its technology capability model, that has synergy with the HE Reference Model.

#03 Build capability (people/roles, responsibilities, process and technology) to support turning data into a high value commodity – data as an asset

#04 Enhance the Enterprise Architecture capability to facilitate transformation of digital services

#05 Adoption of value streams across the operating model will create end-2-end business value across the institutional business domains

Alongside the key observations we have seen specific institutions that have self-assessed their capability areas maturity at **level 3 (Defined)** or above against the CMMI model across the five disciplines. This provides an opportunity for best practise sharing across the wider UCISA community to improve specific capabilities across the sector e.g.:

- **Strategy & Governance** - Operating Model & Tools and Security & Data Protection
- **Sourcing & Optimisation** - Sourcing , Purchasing & Contracts
- **Development** - Requirements & Feasibility
- **Services** - Service and Support of Users, Service Operation & Automation and Service Release and Operational Readiness

Our analysis has revealed that institutions have some way to go on their digital transformation journey with HE sector maturity reducing from the previous reporting period. Additionally, we have shown how the adoption of the BT Standard can provide a foundation for creating digital momentum, set against a framework that encapsulates all areas of the plan, build and run operating model for technology across the business - **it is specifically created to solve the challenge of digital transformation**.

We would like to thank all the institutions that have contributed to the assessment, and we look forward to continuing discussions across the sector. We hope that more institutions will seize the opportunity participate in future maturity assessments, or refresh their existing assessment, that will be performed in collaboration with Xonetic to drive improved insight for the whole community. Please refer to Appendix 2 for further details.

CURRENT STATE DISCIPLINE MATURITY ASSESSMENT OBSERVATIONS

eight DEMAND

CURRENT STATE INSIGHTS / OBSERVATIONS

The Demand Discipline (See Figure 8) provides an average maturity level of **1.96 -Initial** which suggests processes are poorly managed or controlled, there is unpredictable outcomes of processes involved and an ad hoc and chaotic approach used (Lowest quality and highest risk).

The Demand discipline intersects between four other disciplines (strategy and governance, sourcing and optimisation, development and services) and its main objective is to turn business intention into concrete business technology actions, especially in supporting a digital transformation strategy-to-plan and then demand-to-request flow execution.

Demand (PLAN) was seen as the lowest maturity for all institutions against other BT Standard disciplines, with a high degree of respondent disparity across the capability area assessment with a discipline standard deviation (variability of the data values from the mean (average)) of **0.53** across the institutions. **64%** of institutions are at level **1-Initial** maturity level which suggests processes are poorly managed or controlled with unpredictable - Lowest quality and highest risk.

Specific areas of focus from the assessment at level **1-Initial** were on Enterprise Architecture, Data Asset & Integration, Ecosystem Development and Innovation & Concepts capabilities which were reviewed across multiple institutions in the assessment process which supports the low average position where the people, roles, process and/or tools have not been implemented or embedded within the institution to the required level.

We see a high degree of opportunity for improvement across the core Demand discipline in support of creating foundational capabilities for the institutions journey of digital transformation success.

MATURITY ANALYSIS AVERAGES

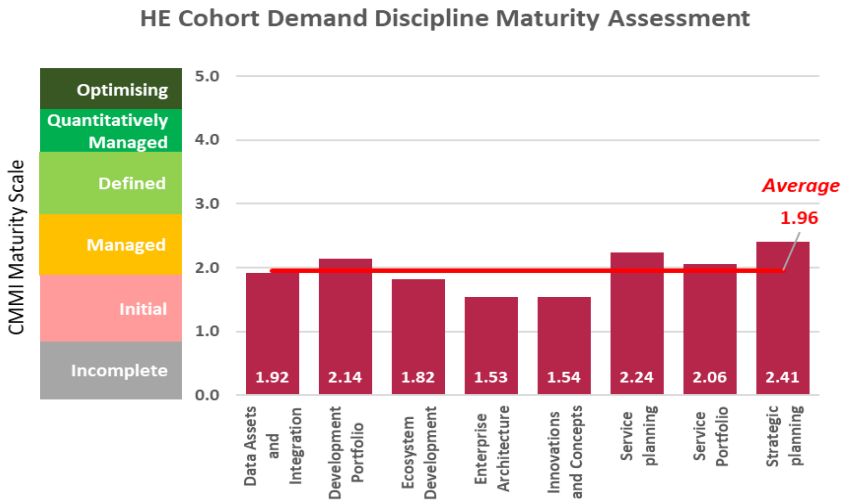


Figure 8 – Demand Discipline Maturity

Demand Discipline	Average CMMI	Standard Deviation
Data Assets and Integration	1.92	0.47
Development Portfolio	2.14	0.84
Ecosystem Development	1.82	0.70
Enterprise Architecture	1.53	0.58
Innovations and Concepts	1.54	0.48
Service planning	2.24	0.60
Service Portfolio	2.06	0.63
Strategic planning	2.41	0.80
Total Average	**1.96**	**0.53**

Please refer to the Business Technology Standard website for more detail on the Demand Discipline

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STRATEGY & GOVERNANCE

MATURITY ANALYSIS AVERAGES

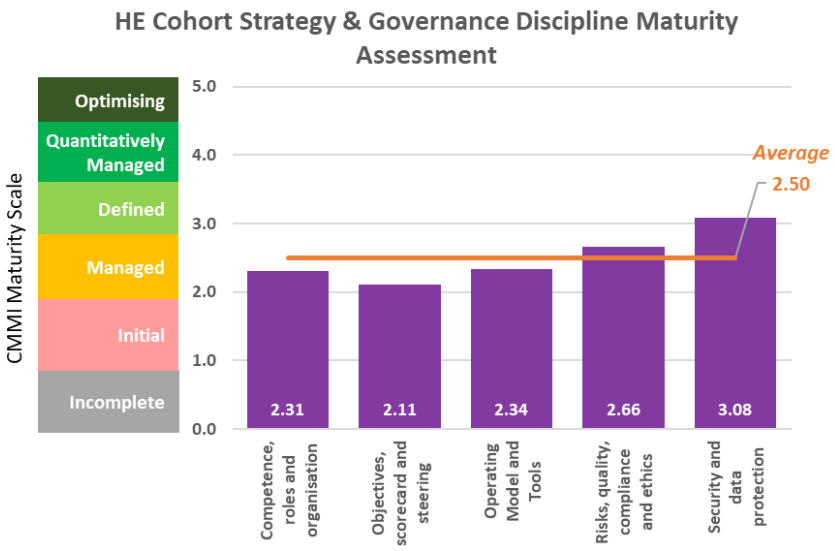


Figure 9 – Strategy & Governance Discipline Maturity

Strategy & Governance Discipline	Average CMMI	Standard Deviation
Competence, roles and organisation	2.31	0.62
Objectives, scorecard and steering	2.11	0.60
Operating models and tools	2.34	0.56
Risks, quality, compliance and ethics	2.66	0.50
Security and data protection	3.08	0.59
Total Average	**2.50**	**0.47**

Please refer to the Business Technology Standard website for more detail on the Strategy & Governance Discipline

CURRENT STATE INSIGHTS / OBSERVATIONS

The Strategy & Governance Discipline (See Figure 9) an average maturity level of **2.50-Managed** which suggests there is planning and controlled implementation of processes and projects across the institutions but there appears to remain a level of reactivity (This risk involved is lower than Initial level, but still exists. Quality is better than Initial level). We have seen a high degree of respondent disparity across the capability area assessment with a discipline standard deviation (variability of the data values from the mean (average)) of **0.47** across the institutions.

The strategy and governance discipline aims to optimise the ways of working for the business technology function by defining the guidelines, rules and framework. The strategy and governance discipline's remit covers these three main topics:

- Set the strategic intention, guidelines and motivation for the organisation
- Implement an operating model and organisation that create value for the business
- Ensure proper management of risks, compliance and security

The Strategy & Governance discipline was seen as the highest maturity for all institutions against other BT Standard disciplines, with Security and Data Protection capability being the highest maturity across most participant institutions with an average of level **3-Defined**, and one institution at level **4-Quantitavely Managed**, which suggests information security is high on the institution's agendas based on legal/regulatory requirements or contractual in support of minimising business risk.

Specific areas of focus from the assessment were on Operating Model & Tools and Objectives, Scorecard & Steering capabilities, which resulted in a deeper dive across multiple institutions in the assessment process. It was seen from respondents that their operating models may not be efficiently defined against plan, build and run to effectively contribute to the creation of business value, using a value stream approach across the business domains, and thus support the transformation for the institution..

ten

SOURCING & OPTIMISATION

MATURITY ANALYSIS AVERAGES

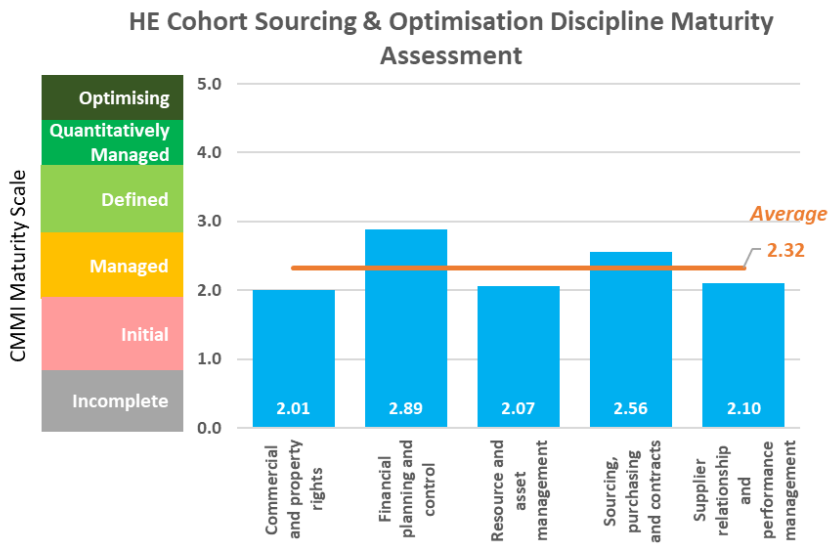


Figure 10 – Sourcing & Optimisation Discipline Maturity

Sourcing & Optimisation Discipline	Average CMMI	Standard Deviation
Commercial and property rights	2.01	0.63
Financial planning and control	2.89	0.64
Resource and asset management	2.07	0.56
Sourcing, purchasing and contracts	2.56	0.69
Supplier relationship and performance management	2.10	0.58
Total Average	**2.32**	**0.52**

Please refer to the Business Technology Standard website for more detail on the Sourcing & Optimisation Discipline

CURRENT STATE INSIGHTS / OBSERVATIONS

The Sourcing & Optimisation Discipline (See Figure 10) provides an average maturity level of **2.32-Managed** which against suggests there is planning and controlled implementation of processes and projects across the institutions but there appears to remain a level of reactivity (This risk involved is lower than Initial level, but still exists. Quality is better than Initial level). Again, we saw a high degree of respondent disparity with a discipline standard deviation (variability of the data values from the mean (average)) of **0.52** across the institutions.

The Sourcing & Optimisation discipline is an end-to-end discipline that manages the whole lifecycle of business services and vendor relations. It implements the strategy and the operating model by engaging vendors, contracts and resources to deliver high quality products, solutions and services at appropriate cost levels.

Financial Planning & Control capability was seen as a stronger area of maturity at level **2-Managed** (level **3-Defined** in April 2022 report) which suggests as semi effective management of business technology with analytical financial steering and cost allocation approaches, albeit still with a level of reactivity in this approach across the business domains in support of operational performance and investment feasibility.

Specific areas of focus from the assessment were on Resource & Asset Management capability which resulted in a deeper dive across multiple institutions. It was shown from the data responses that there was limited processes against Service Request Management/Fulfilment, aligned to a Service Catalogue of standard/normal resources and technology assets available for the institution, and the technology ecosystem to support the SMART sourcing of resources. There is also a further option of a self-service portal – shift left capability. This issue is linked to the potential lack of run tooling that supports both Demand and Services disciplines with automation/AI capabilities and use of data analytics to drive improved demand forecasting from historical demand data. This could be an opportunity across the sector to develop a standard approach.

eleven DEVELOPMENT

CURRENT STATE INSIGHTS / OBSERVATIONS

The Development Discipline (See Figure 11) provides an average maturity level of **2.21-Managed** which suggests there is planning and controlled implementation of processes and projects across the institutions but there appears to remain a level of reactivity (This risk involved is lower than Initial level, but still exists. Quality is better than Initial level). We again saw a high degree of disparity with a discipline standard deviation (variability of the data values from the mean (average)) of **0.62** across the institutions.

The Development (BUILD) discipline supports the enablement of information technology that is planned, built, operated and continuously. The majority of an organisation's development initiatives are based on existing solutions and services. In order to transform and expand a business, innovative solutions and services are a necessary addition. Development initiatives should be organised into value streams, which guide the understanding, organising, and value delivery for a business.

Requirements & Feasibility capability was seen as the strongest area of maturity, although still at level **2-Managed** across the participants, which suggests that institutions use of sequential and incremental development is supported by a baseline set of agreed requirements before the actual development starts. This is an essential part of any development request and the foundation upon which a solution, product or service should be built.

A specific area of focus from the assessment was on Business Value Realisation capability (level **1-Initial**) and how it sits within their Development framework/life cycle - from Demand (Innovation to Concepts) through to benefits/value realisation aligned against the defined value streams of the institutions operating model.

MATURITY ANALYSIS AVERAGES

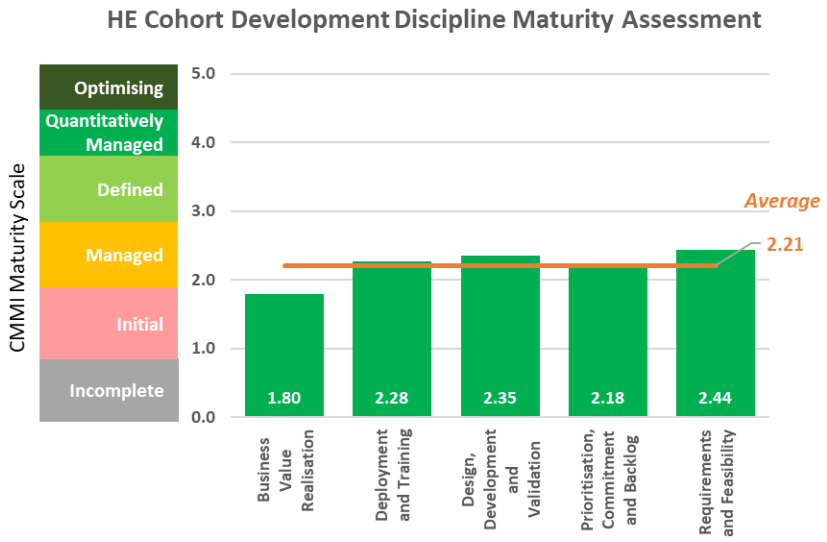


Figure 11– Development Discipline Maturity

Development Discipline	Average CMMI	Standard Deviation
Business Value Realisation	1.80	0.65
Deployment and Training	2.28	0.50
Design, Development and Validation	2.35	0.73
Prioritisation, Commitment and Backlog	2.18	0.77
Requirements and Feasibility	2.44	0.74
Total Average	**2.21**	**0.62**

Please refer to the Business Technology Standard website for more detail on the Development Discipline

twelve

SERVICES

MATURITY ANALYSIS AVERAGES

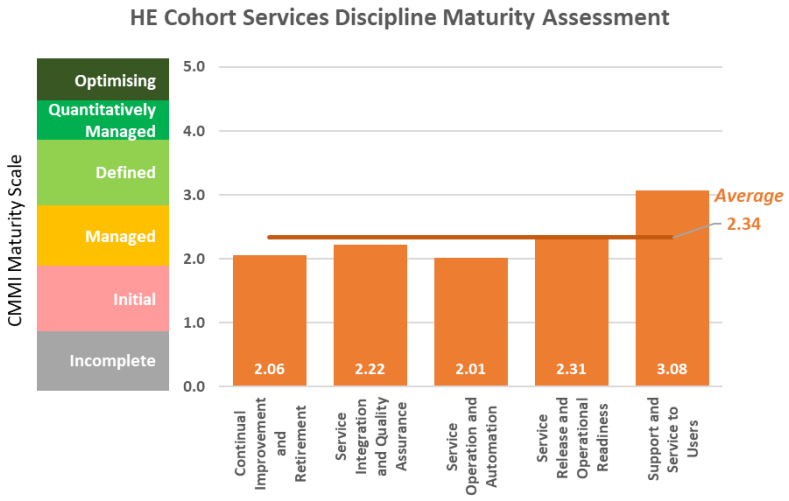


Figure 12 – Services Discipline Maturity

Services Discipline	Average CMMI	Standard Deviation
Continual Improvement and Retirement	2.06	0.59
Service Integration and Quality Assurance	2.22	0.67
Service Operation and Automation	2.01	0.64
Service Release and Operational Readiness	2.31	0.70
Support and Service to Users	3.08	0.71
Total Average	**2.34**	**0.57**

Please refer to the Business Technology Standard website for more detail on the Services Discipline

CURRENT STATE INSIGHTS / OBSERVATIONS

The Services Discipline (See Figure 12) provides an average maturity level of **2.34-Managed** which suggests there is planning and controlled implementation of processes and projects across the institutions but there appears to remain a level of reactivity (This risk involved is lower than Initial level, but still exists. Quality is better than Initial level). Again, we saw a high degree of disparity with a discipline standard deviation (variability of the data values from the mean (average)) of **0.57** across the institutions.

The Services (RUN) discipline makes the tested and validated development outcomes accessible to users via a service release. Services discipline ensures the operational readiness with the service delivery ecosystem and validates that user support and services are in place with required skills and capacity.

The Support and Service to Users capability was the strongest maturity with 62% of institutions scoring themselves at level **3-Defined** and up to level **4-Quantifitavely Managed** which suggests that Service (Run) capability is in place that supports responses to day-to-day questions, handle service requests and resolves incidents from academics, staff and students, as well as possible external partners.

There is still room for improvement across institutions and the we have already identified the potential lack of run based tooling, that supports both Demand and Services disciplines with service operation and automation/AI capabilities, as one such opportunity across the sector to develop to further optimise the discipline.

A specific areas of focus from the assessment was on the Service Operations and Automation and Continual Improvement and Retirement capability with **36%** of institutions at level **1-Initial** maturity level or lower. This suggests that the processes are poorly managed or controlled which creates unpredictable outcomes - low quality and higher risk.

APPENDICIES

appx1

CAPABILITY MATURITY MODEL INTEGRATION (CMMI)

5	Optimising	The capability area is managed with continued improved capability development through incremental & innovative technology improvements
4	Quantitatively Managed	The capability area is managed & controlled using defined quantitative indicators and techniques
3	Defined	The capability area has been implemented, understood, well-defined and measured. Proactively managed
2	Managed	The capability area is planned but no controlled implementation. Often reactively managed
1	Initial	The capability area in unpredictable, poorly controlled and relative
0	Incomplete	The capability area is undefined or incomplete. The capability is performed occasionally at best

Figure 13 – CMMI maturity Scale

The Capability Maturity Model Integration (CMMI) was developed by Carnegie Mellon University and its goal was to make maturity models - which measure the ability of organisations to have ongoing improvement in a particular area - more effective and usable by integrating a number of models into a single framework.

The CMMI model that can help organisations to effect process improvement and develop behaviours that decrease risks in service, product and development.

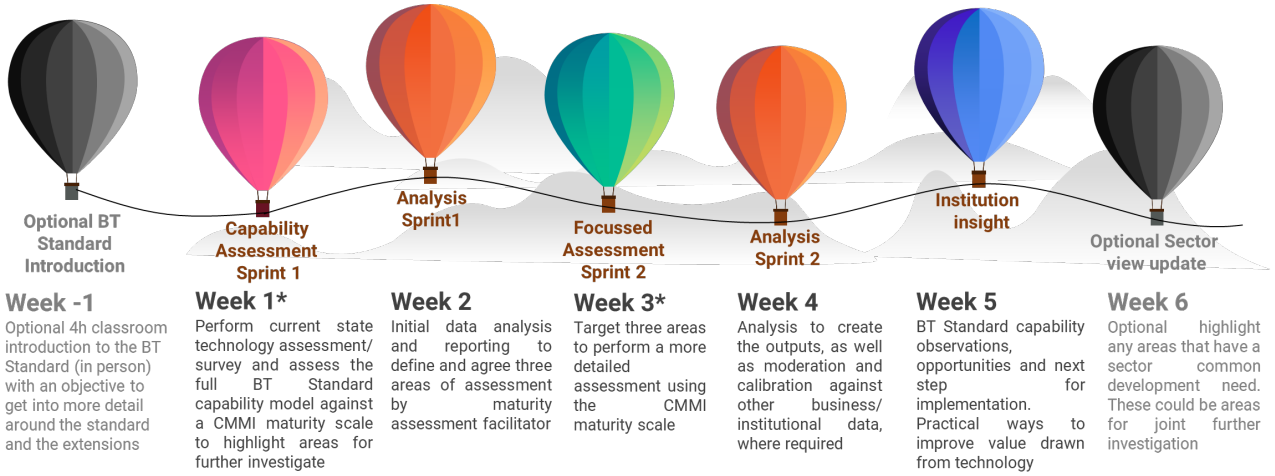
The model enables organisations to measure, build, and improve capabilities - to improve overall performance.

BT STANDARD MATURITY ASSESSMENT

How does the IT Maturity Assessment Work?

- It is carried out using a virtual collaboration tool known as **Howspace**, which combines facilitation expertise with business social media in an agile, fun and inspirational way
- Assessment of IT performed against the BT Standard Capability Model – 5 disciplines and 28 standardised capability areas
- Participants who are knowledgeable in the way technology is used within your institution contribute in short, intuitive and timed assignments of 5 days sprints ie: 15-minute interaction time, per day, should be planned for or contributors can choose to perform longer interaction sessions, over fewer days
- **(Optional) BT Standard Introduction** (Week -1) can provide a classroom introduction to the BT Standard (in person) with the objective to get into more detail around the standard and the extensions in preparation for Sprint 1
- **Sprint 1** (Week 1) provides high-level assessment of IT across the 28 standardised capability areas using an industry standard maturity scale (see Appendix 1)
- **Sprint 2** (Week 3) will perform a more focus review of upto 3 specific capability areas based on the outputs from Sprint 1
- Once sprint collaboration sessions have been run to collect information about your institution, Xonetic experts will assess and analyse the data to drive observational reporting and sector assessment through Weeks 4 - 6

Maturity Assessment Timeline



Costs

Market rate £20,000 + VAT

Optional BT Standard Introduction discounted to £800 + VAT for UCISA members

New maturity assesments discounted to £4,999 + VAT for UCISA members

Refresh maturity assessments discounted to £2,500 + VAT for UCISA members

Please note

- Costs are based on a cohort of **four (4)** universities working together for a new assessment. Should the total number exceed four, a second cohort can be scheduled. Please note that should the number of institions in the cohort be less than four Xonetic reserve the right to amend the individual institution cost.
- Costs for refresh assessments are individual and are based on a single Sprint 1 assessment and report across the BT Standard Capability Model. Recommendation is that refresh is performed annually
- All work will be performed remotely.
- Please contact Nick Russell of Xonentic (Nick.Russell@Xonetic.com) for further details of maturity assessment process and next scheduled cohort.

BT STANDARD MATURITY ASSESSMENT NEXT STEPS

The simplest next step for institutions who have completed the BT Standard maturity assessment is to ensure broad and equal understanding of standard in an easy to consume way, provided by an expert to allow more detailed discussion and alignment on institution priorities and opportunities

BT Standard Green Card

Length and method: 1h online test
The quiz consists of 40 multiple-choice questions.

Objective: Getting familiar with the Business Technology Standard
Target group: Anyone wanting to test and evidence understanding the basics of the Business Technology Standard

£FREE

BUSINESS TECHNOLOGY ACADEMY

Classroom introduction

Length and method: 4h classroom introduction to the BT Standard – in person

Objective: Get into more detail around the standard and the extensions.
Target group: Anyone wanting to further understand the Business Technology Standard, and implementation advantages

£800 plus expenses + VAT

- Run by a single Xonetic Resource
- Note taking supported by institution

Workshop discussion

Length and method: Homework from previous session and then 4h workshop – in person

Objective: Discuss the specifics of how institution could implement the standard, where/how to start, using the maturity assessment output as a guide. Agree the action plan.

Target group: Institution Leadership team

£800 plus expenses + VAT

- Run by a single Xonetic Resource
- Note taking supported by institution
- ***Optional***: Write up of findings and production of approach document 2h - £250 + VAT

appx 3

WHO ARE XONETIC

We Innovate, We Care and We have Fun!

We are an independent **Business Technology** consultancy that helps businesses make the best use of technology to build **Digital Energy.** We help organisations combine the latest **Cloud technologies** (products) with **modern ways of working** (Business Technology), so that they can enhance business efficiency and performance.

Our core markets include Financial Services, Retail and Higher Education.

We believe that along with access to the latest technology and innovation, businesses need to embrace new digital ways of working. This creates a pathway for the full use of technologies such as **AI, robotics and data analytics**.

We have continued to develop our internal associate family and have retained a strong partnership with Sofigate, which provides us with access to over **600 consultants** across the Nordics and **ServiceNow/RPA** capacity in Finland.

We are small in size, so we are able to focus on quality and value. Thereby enabling us to curate customised solutions for our clients.

2015
3Gamma

Our Nordic heritage begins.

3Gamma, a Swedish Management Consultancy creates 3Gamma UK.

2016
Strategy Begins

Nick Russell joins the UK business to establish Strategy Consulting.

2017
Sofigate

3Gamma is aquired by Sofigate, a Finnish Technology Management Specialist.

2019
Xonetic

Sofigate UK is acquired by Nick Russell and becomes an independent business, whilst maintaining close ties with Sofigate.

2023+
Future

Xonetic continues to spread the message of Digital Energy and Business Technology to old and new customers alike.

RESPONDENT INSTITUTIONS

Institution	Date of Participation	Respondents		
		IT	Business	Total
King's College London	07/02/2022	4	1	**5**
Leeds Trinity University	12/06/2023	10	4	**14**
London School of Hygiene & Tropical Medicine	26/09/2022	5	3	**8**
Nottingham Trent University	07/02/2022	6	0	**6**
Queens Belfast University	26/09/2022	11	5	**16**
Queen Mary University of London	07/02/2022	3	1	**4**
University of Glasgow	07/02/2022	4	1	**5**
University of Huddersfield	04/07/2022	6	7	**13**
University of Hull	23/05/2022	6	5	**11**
University of Oxford	23/05/2022	3	5	**8**
University of St Andrews	07/02/2022	3	2	**5**
University of Surrey	04/07/2022	5	5	**10**
University of Winchester	04/07/2022	5	7	**12**
University of York	07/02/2022	4	2	**6**

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